

Reliable Donation-Commitment Labeling and Classification for PersuasionForGood

Arnob Biswas, Nawriz Ahmed Turjo, Abhishek Roy, Monjur Hossain Khan, Shams Hossain Simanto

Dept. of Computer Science and Engineering

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

Abstract—The PersuasionForGood corpus pairs each of its 1,017 charitable-donation dialogues with a numeric outcome field recording whether, and how much, the Persuadee donated – but this field is demonstrably unreliable: in multiple dialogues it disagrees with what the Persuadee actually says in the transcript (e.g., a \$0 field on a dialogue where the Persuadee explicitly confirms a \$10 donation). This project builds a manually annotated, text-grounded replacement: each dialogue is labeled with a binary commitment field (`binary_label`: yes/no – did the Persuadee make an unconditional donation commitment) and a modifier (`none/conditional/deferred`) that captures commitments the original binary/numeric fields cannot represent at all. Five annotators are each labeling an independent partition of roughly 200 dialogues toward a combined target of 1,000 labeled dialogues; this report covers the lead partition, currently 204 dialogues (161 yes, 43 no; modifiers: 186 none, 17 deferred, 1 conditional). An independent second annotation pass, run over an initial 154-dialogue slice of this partition with no access to the first pass’s labels, agreed on 142/154 dialogues (92.21%), and the resulting disagreement analysis directly shaped the annotation guidelines used to label the remaining dialogues. This report presents the motivating gap in the corpus’s own outcome field, the annotation scheme and its reliability so far, and the planned classifier that will predict `binary_label/modifier` directly from dialogue text once the full 1,000-dialogue set is complete.

Index Terms—donation-intent labeling, ground-truth annotation, annotation reliability, dialogue outcome classification, PersuasionForGood

I. INTRODUCTION

Persuasive dialogue systems for charitable donation are usually evaluated against the PersuasionForGood corpus [1], which pairs 1,017 human–human donation-persuasion dialogues with a numeric outcome field recording whether, and how much, the Persuadee donated. That field is treated as ground truth by essentially every downstream analysis built on the corpus, including the most recent and most thorough one, Petrova et al.’s 41-strategy re-annotation of all 10,600 persuader turns [2], which uses it directly to test whether persuasion strategies predict donation outcome.

The problem is that the field is not reliable. Cross-checking it against the dialogue text itself surfaces clear disagreements: dialogues where the Persuadee explicitly and unambiguously confirms a donation amount in the chat while the corpus’s own field records \$0, and dialogues where a hedged or conditional statement (“I’ll think about it,” “if I look them up and they check out, sure”) gets collapsed into the same binary bucket as

an unconditional yes. Petrova et al. report that single-strategy-category presence explains almost none of the variance in this outcome field (pseudo- $R^2 \approx 0.015$ – 0.08); it is an open question how much of that weak signal is a genuine null result about persuasion strategy, versus noise inherited from an outcome label that does not reliably reflect what the Persuadee actually said.

This project addresses that gap directly: build a manually annotated, text-grounded donation-commitment label for each PersuasionForGood dialogue, validate its reliability, and use it to train a classifier that predicts commitment status directly from the dialogue transcript. The label scheme is a binary commitment field (`binary_label`: did the Persuadee make an unconditional donation commitment) plus a three-way modifier distinguishing an unqualified commitment (`none`) from one that is `conditional` (contingent on something the Persuader never confirms) or `deferred` (genuine intent pushed outside the current task) – a distinction the corpus’s original binary/numeric fields cannot represent at all. Section II states the annotation and classification problem more precisely; Section III positions this project against prior PersuasionForGood work; Section IV describes the corpus and the annotation dataset; Section V lays out the annotation-then-classification pipeline; Section VI reports preliminary annotation and reliability results; and Section VIII lists the work remaining for the project.

II. PROBLEM DESCRIPTION

The central question is whether a manual annotation effort can produce a donation-commitment label that is both (a) more faithful to what the Persuadee actually said than the corpus’s own outcome field, and (b) learnable by a text classifier well enough to serve as a scalable replacement for that field. Framing this as a genuine, non-trivial problem rests on several compounding difficulties:

- **The source field disagrees with the source text.** Spot-checking the corpus’s numeric outcome field against the dialogue transcript surfaces cases where the two are flatly inconsistent (e.g., a recorded \$0 donation despite an explicit \$10 confirmation in the chat). A replacement label has to be built directly from the text, not inherited from the field it is meant to correct.
- **Binary is too coarse.** A plain yes/no field cannot distinguish an unqualified commitment from one that is

conditional on something the Persuader never confirms, or one that is genuinely intended but deferred outside the current task – distinctions that matter both for accurately describing what happened in the dialogue and for any downstream outcome analysis. This motivates the `modifier` field alongside `binary_label`.

- **Annotation reliability is not free.** The reliability pilot reported in Section VI reaches 92.21% two-pass agreement on 154 dialogues, but that check is within a single annotator’s own partition (one annotator, double-passed). Consistency across the five independently labeled partitions is instead enforced by an LLM cross-check rather than a dedicated pairwise human inter-annotator study: once a partition is fully labeled, an LLM independently re-labels the same dialogues, and every row where the LLM’s label disagrees with the human annotator’s is flagged and re-adjudicated by the rest of the annotation team – not the original annotator – before the partition is finalized. This catches partition-specific drift more cheaply than a full second human pass would, but it is only as reliable as the LLM’s own labeling behavior on this task, which has not itself been separately validated.
- **Class imbalance threatens the classifier stage.** In the lead partition, `binary_label` is skewed 161 `yes` to 43 `no` (78.9%/21.1%), and `modifier` is heavily dominated by `none` (186/204), with only 17 `deferred` and a single `conditional` example so far. A classifier trained naively on this distribution risks collapsing to the majority class, particularly for `modifier`.

III. STATE OF THE ART

Foundational corpus and its outcome field. Wang et al. [1] introduced PersuasionForGood together with the numeric donation-outcome field that nearly every downstream analysis of the corpus – most recently Petrova et al. [2] – treats as ground truth without re-checking it against the dialogue text.

Strategy classifiers built on the corpus. Wang et al.’s original 10-strategy single-label classifier and Saha et al.’s [3] later BERT-based re-implementation both predict persuader strategy, not the donation-outcome label itself. Petrova et al. [2] extend this to a 41-strategy, 11-category taxonomy and use it, alongside the same unverified outcome field, to test strategy-outcome correlation – finding a notably weak one (pseudo- $R^2 \approx 0.015$ – 0.08). None of this line of work questions the outcome field itself.

Persuadee-side modeling. Tian et al. [4] annotate the Persuadee’s resistance strategies on the same corpus – evidence that persuadee-side labels beyond the corpus’s original outcome field are both feasible to collect and informative, a precedent this project follows for a different persuadee-side label (commitment status rather than resistance strategy).

Outcome and susceptibility prediction elsewhere. A 2025 framework (DASS) [5] predicts which persuasive strategies a Persuadee is susceptible to, on a re-annotated PersuasionForGood-derived dataset (P4G+) – evidence that persuadee-side outcome labels on this corpus can be both

TABLE I
WHAT EXISTS VS. WHAT THIS PROJECT CONTRIBUTES

Already exists	Still missing (this project)
A numeric donation-outcome field bundled with the corpus, used as ground truth by every downstream study (2019–2026)	Manual verification of that field against the dialogue text, and a finer-grained yes/no + conditional/deferred replacement
Single-label persuader-strategy classifiers (2019, 2021, 2026)	A classifier trained to predict the corrected commitment label directly from dialogue text
Persuadee-side resistance-strategy annotation on this corpus (2020)	The same persuadee-side annotation discipline applied to commitment status
Weak strategy-outcome correlation reported using the uncorrected field (2026)	Re-testing that relationship, if time permits, once a validated outcome label exists

re-annotated and learned. Chen and Yang [6] show that persuasive-strategy labels can be learned from partial, weakly supervised annotation on good-faith textual requests, relevant if manual annotation cannot fully cover the 1,000-dialogue target on schedule. Chen and Xiao [7] apply multi-label classical ML methods (random forests, classifier chains) to a related persuasion-labeling problem outside PersuasionForGood; useful as a modeling reference for the classifier stage of this project even though their label space, domain, and (single-vs. multi-label) task structure differ from ours.

Positioning. Table I summarizes the gap.

IV. DATA

This project uses the existing, publicly released PersuasionForGood corpus [1] as its dialogue source – no new dialogue collection is planned. The corpus contains 1,017 human–human dialogues collected on Amazon Mechanical Turk, in which one worker (the Persuader) attempts to convince the other (the Persuadee) to donate part of their task payment to the charity Save the Children; it comprises 10,600 persuader turns and 10,332 persuadee turns, released with per-turn text (`conversations.json`, `utterances.jsonl`) alongside the numeric outcome fields discussed in Section II.

Annotation scheme. On top of the raw corpus, this project builds a manually produced donation-commitment ground truth. Each row records five fields: `conversation_id`, `dialogue_id`, the full `dialogue_text`, `binary_label` (yes/no – did the Persuadee make an unconditional commitment to donate), and `modifier` (none/conditional/deferred), built to correct cases where the corpus’s own numeric outcome fields do not match what the Persuadee actually says in the dialogue text – for example, one conversation records a \$0 donation even though the Persuadee explicitly confirms a \$10 donation in the chat.

Annotation partitions. Five annotators are each independently labeling a partition of roughly 200 dialogues, toward a combined target of $5 \times 200 \approx 1,000$ labeled dialogues. This report covers progress on the lead partition

TABLE II
LEAD-PARTITION ANNOTATION PROGRESS (204 DIALOGUES LABELED)

Field	Value	Count (%)
binary_label	yes	161 (78.9%)
	no	43 (21.1%)
modifier	none	186 (91.2%)
	deferred	17 (8.3%)
	conditional	1 (0.5%)

(ManualLabel.csv); the other four partitions follow the identical schema and are in progress.

Both fields are visibly imbalanced at this stage – most notably modifier, where conditional has only a single labeled example so far. Whether this reflects the true rarity of conditional commitments in the corpus or is a lead-partition sampling artifact will become clearer once the remaining four partitions are complete (Section VIII); either way, it is a concrete risk factor for the classifier stage of Methods (Section V).

V. METHODS

The planned pipeline has three stages.

1) Manual annotation. Each of five annotators independently labels a ~ 200 -dialogue partition of Persuasion-ForGood with (binary_label, modifier), following a shared guideline document that is iteratively refined from the disagreement patterns surfaced during reliability checks (Section VI). This stage is already under way: the lead partition stands at 204 of its ~ 200 -dialogue target, and the two-pass reliability pilot on that partition (Section VI) is what produced the guidelines the remaining four partitions are being labeled against. Once a partition is fully labeled, it goes through an LLM cross-check (Section II): the LLM independently labels the same dialogues, and every row where its label disagrees with the human annotator’s is flagged and re-adjudicated by the rest of the annotation team rather than by the original annotator. Once all five partitions have been labeled and passed through this cross-check, they are pooled into the combined $\sim 1,000$ -dialogue dataset.

2) Classifier training. Fine-tune a text encoder (e.g., RoBERTa-base or DeBERTa-v3-base) with two output heads: a binary head for binary_label and a three-way head for modifier (conditioned on binary_label = yes), trained on the pooled partitions with a train/validation/test split stratified by partition to check for partition-specific label drift. Evaluation: accuracy and F1 for the binary head, macro-F1 for the modifier head, and a majority-class baseline given the class imbalance already observed (Table II).

3) Comparison against the original field. Compare the corrected binary_label against the corpus’s original numeric outcome field across the full $\sim 1,000$ -dialogue set, to quantify – beyond the spot-checked examples in Section IV – how often, and in what patterns, the two actually disagree.

VI. PRELIMINARY RESULTS

As the first stage of Methods Step 1 (Section V), the lead partition currently stands at 204 dialogues labeled with the binary_label/modifier scheme (Table II). An independent second annotation pass was run over the initial 154 dialogues of this partition, labeling strictly from the dialogue text and guidelines with no access to the first pass’s labels (to avoid anchoring).

Reliability check. The two passes agreed on 142/154 dialogues (**92.21%**); 12 dialogues (7.79%) disagreed, and a further 28 borderline calls were flagged across the batch as cases a different labeler could reasonably call differently even where the two passes happened to agree.

Disagreement patterns. The 12 disagreements cluster into a small number of recurring boundary cases rather than being spread randomly across the guideline:

- *Hedged-but-unretracted phrasing* (“I may give a few cents,” “maybe I will consider it”) – the two passes differed on whether a hedge with no explicit retraction counts as a commitment.
- *The conditional/deferred boundary* – e.g., a Persuadee who says “if I research them, and they are doing good things, then yes” was labeled deferred by one pass and conditional by the other, since the condition (research) is the Persuadee’s own future action rather than something the Persuader must confirm.
- *Persuader match-offers* – whether a Persuadee’s brief agreement to a Persuader’s “I’ll match your donation” constitutes a firm yes or an unresolved exchange.
- *Off-topic or malformed task instances* – one dialogue never reaches an actual donation ask, and another has the Persuadee explicitly stating they were “just role playing,” both of which produced different calls across passes.

These patterns directly informed a “recurring label calls” section added to the annotation guidelines (e.g., codifying that hedged-but-unretracted amounts count as yes, while pure hedges with no follow-through count as no) before the remaining 50 dialogues of the lead partition were labeled, bringing it to its current 204. The same guidelines are being used by the other four annotators as they label their own partitions toward the combined 1,000-dialogue target.

VII. DISCUSSION

The 92.21% two-pass agreement rate is encouraging in absolute terms, but the more useful signal is where the 7.79% of disagreements concentrate: not randomly, but in a handful of linguistically identifiable boundary categories (hedged commitments, the conditional/deferred distinction, match-offer resolution). That concentration matters for the rest of this project in two ways.

First, it suggests annotation guidelines can be iteratively tightened against a small set of recurring failure modes rather than needing open-ended rewriting – exactly the process used here between the first and second annotation passes on the lead partition, and the process the guidelines are now carrying into the other four partitions.

Second, the label imbalance in Table II – particularly `modifier`’s single conditional example against 186 `none` – is a concrete risk surfaced by the preliminary data rather than a theoretical concern: if the remaining four partitions show a similarly skewed distribution, the planned three-way `modifier` classifier (Methods Step 2) may need class-weighted loss, oversampling of the minority classes, or a coarser `none-vs.-qualified` collapse to be trainable at all.

VIII. PLANNED WORK

The remaining work follows a tentative five-week plan against the three Methods stages (Section V) – subject to confirming group availability:

- 1) **Week 1 (complete).** Manual annotation of the lead partition: 204 dialogues labeled with the `binary_label/modifier` scheme, including the two-pass reliability pilot (Section VI) that produced the current annotation guidelines.
- 2) **Week 2.** Complete manual annotation on the remaining four partitions, bringing the combined dataset from 204 dialogues toward the $\sim 1,000$ -dialogue target, using the guidelines refined in Week 1; run each newly completed partition through the LLM cross-check described in Methods Step 1 (independent LLM labeling, followed by team re-adjudication of any human/LLM mismatches); then consolidate all five partitions into a single pooled dataset and finalize it for training: schema checks, dedup, and a train/validation/test split stratified by partition.
- 3) **Weeks 3–4.** Classifier training and evaluation (Methods Step 2): fine-tune the two-head text classifier on the pooled dataset; report binary-head F1, modifier-head macro-F1, and the majority-class baseline for comparison. Given two heads to train and tune, this stage is allotted two weeks rather than one.
- 4) **Week 5.** Comparison against the original outcome field (Methods Step 3): quantify how often the corrected `binary_label` disagrees with the corpus’s original numeric outcome field across the full dataset, and write up the final report.

IX. CONCLUSION

The PersuasionForGood corpus’s numeric donation-outcome field is treated as ground truth by every downstream analysis built on it, including the most recent and most thorough strategy re-annotation of the corpus, yet it visibly disagrees with the dialogue text in identifiable cases. This project builds a manually annotated, text-grounded replacement – a binary commitment label plus a conditional/deferred modifier – and reports preliminary progress: 204 dialogues labeled in the lead partition of a planned 1,000-dialogue, five-annotator dataset, with 92.21% within-annotator two-pass agreement and a small, identifiable set of boundary categories driving most disagreement. The remaining work – completing annotation across all five partitions, training a classifier to predict the corrected label

from dialogue text, and comparing it against the corpus’s original outcome field – is scoped as a five-week plan in Section VIII, of which the first week is already complete.

REFERENCES

- [1] X. Wang, W. Shi, R. Kim, Y. Oh, S. Yang, J. Zhang, and Z. Yu, “Persuasion for good: Towards a personalized persuasive dialogue system for social good,” in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019. [Online]. Available: <https://arxiv.org/abs/1906.06725>
- [2] A. Petrova, K. Sokol, and R. State, “How much does persuasion strategy matter? llm-annotated evidence from charitable donation dialogues,” *arXiv preprint arXiv:2604.19783*, 2026. [Online]. Available: <https://arxiv.org/abs/2604.19783>
- [3] S. Saha, K. Kalra, M. Patwardhan, and S. Karande, “Performance of BERT on Persuasion for Good,” in *Proceedings of the 18th International Conference on Natural Language Processing (ICON)*, Silchar, India, 2021.
- [4] Y. Tian, W. Shi, C. Li, and Z. Yu, “Understanding user resistance strategies in persuasive conversations,” in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020. [Online]. Available: <https://aclanthology.org/2020.findings-emnlp.431/>
- [5] Anonymous, “Towards strategic persuasion: Unveiling users’ susceptibility to persuasive strategies in dialogues,” *Knowledge-Based Systems*, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S0950705125020374>
- [6] J. Chen and D. Yang, “Weakly-supervised hierarchical models for predicting persuasive strategies in good-faith textual requests,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, 2021. [Online]. Available: <https://arxiv.org/abs/2101.06351>
- [7] S. Chen and L. Xiao, “Predicting and characterising persuasion strategies in misinformation content over social media based on the multi-label classification approach,” *Journal of Information Science*, 2025. [Online]. Available: <https://journals.sagepub.com/doi/abs/10.1177/01655515231169949>